

Effective (Spur-Free) Dithering of Digital Delta-Sigma Modulators with Pseudorandom Dither

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Abstract—Digital delta-sigma modulators (DDSMs) are finite state machines; their spectra are characterized by strong periodic tones (so-called spurs) when they cycle repeatedly in time through a small number of states. This can happen for a range of constant and periodic inputs. Pseudorandom dither generators are widely used to break up periodic cycles in DDSMs in order to eliminate spurs produced by underlying periodic behavior. Unfortunately, pseudorandom dither signals are themselves periodic and therefore can produce spurs in the output spectrum. We call pseudorandom dithering effective if it breaks up periodic cycles in the modulator *without* producing its own spurs in the output spectrum. This paper presents rigorous mathematical analysis and a design methodology to achieve effective dithering.

I. INTRODUCTION

Digital delta-sigma modulators (DDSMs) are key building blocks in a wide range of modern communications subsystems such as fractional- N frequency synthesizers. In a DDSM, a high resolution discrete-time input is oversampled and re-quantized to produce a lower resolution output. This coarse requantization takes place within a feedback loop such that the resulting quantization noise power is attenuated in the signal band of interest.

A DDSM is a Finite State Machine (FSM). Consequently, it produces a periodic output (cycle) when the input is constant or periodic and the resulting quantization “noise” behaves in a periodic manner. A discrete-time periodic signal with period L_s is characterized by a frequency spectrum which contains components at DC and integer multiples of the fundamental frequency $\frac{f_s}{L_s}$, where f_s is the clock frequency. When short cycles are produced in a DDSM, this results in strong frequency components called “spurious tones” (spurs).

Over the past decade, much attention has been paid to reducing the magnitudes of spurs introduced by DDSMs. Two classes of techniques have been developed: (a) deterministic, which attempt to spread the energy of the quantization noise signal over as many tones as possible by maximizing the period L_s , and (b) stochastic, which break up the inherent periodicity of the signal by adding a random dither signal. State of the art deterministic techniques include setting predefined inputs and/or initial conditions [1]–[3], using prime modulus quantizers [4], and architectural modifications [4], [5].

The most popular *stochastic* technique is the use of additive least significant bit (LSB) dithering [6], [7] which can be introduced at various points of the DDSM architecture to break up periodic cycles [8]. In principle, the random additive LSB dither signal is non-periodic and the resulting phase

noise spectrum of the DDSM is smooth. In practice, however, the stochastic technique uses a dither signal which is not random; rather, it is usually produced by another finite state machine [9]. The most common approach is to use as dither the signal produced by a pseudorandom binary sequence (PRBS) generator; note that this is a *periodic* signal.

Intuitively, one might expect that the period of the pseudorandom binary sequence should be as *large* as possible to make the dither signal random. However, a recent paper suggests that a relatively *small* 8-bit linear feedback shift register (LFSR) dither generator is especially “efficient” at eliminating spurs [8]. We have highlighted a major drawback of this approach, namely that a dither generator based on a small LFSR introduces spurs due to its inherent periodicity. To illustrate the problem, we have previously considered a MASH architecture which incorporates a dither strategy using LFSR dither generators with different wordlengths [10], [11]. In this paper we present a theoretical framework and design methodology which produces an optimum LFSR wordlength through codesign of the DDSM and a single LFSR.

This paper is organized as follows. In Section II, we review the generic MASH DDSM architecture. In Sec. III, we show how to calculate the heights of the spurs due to periodic dither. In Sec. IV, we show how to hide the spurs due to the periodic dither under the DDSM quantization noise. Representative examples are presented in Sec. V. We discuss our observations and summarize our conclusions in Sec. VI.

II. MASH ARCHITECTURE

The basic building block of a typical MASH DDSM is the first order error feedback modulator (EFM1) shown in Fig. 1. The input to the modulator is a digital word with N bits. When $v[n]$ is greater than the modulus M (where $M = 2^N$ in this case), the quantizer overflows and the output signal $y[n]$ (the carry out) will be 1. On the other hand, when $v[n]$ is less than M , the quantizer does not overflow and $y[n]$ will be 0. Mathematically, we write:

$$y[n] = \begin{cases} 0, & v[n] < M, \\ 1, & v[n] \geq M, \end{cases} \quad (1)$$

Fig. 2 shows a block diagram of an l^{th} order MASH DDSM comprising a cascade of N -bit EFM1 blocks and an error cancellation network. In this structure, the negative of the quantization error from each stage ($-e_i[n]$) is passed to the next stage and the outputs of the stages ($y_i[n]$) are combined in

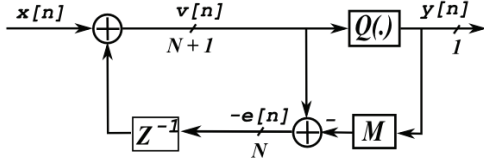


Fig. 1. Block diagram of a first-order error feedback modulator (EFM1)

an error cancellation network, the role of which is to eliminate the intermediate quantization noise terms.

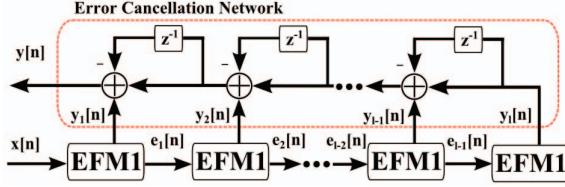


Fig. 2. Block diagram of an l^{th} order MASH DDSM incorporating a cascade of EFM1s

The output of the l^{th} order MASH DDSM can be expressed in the Z-domain as:

$$Y(z) = \frac{1}{M}X(z) + \frac{1}{M}(1 - z^{-1})^l E_l(z). \quad (2)$$

where $X(z)$ and $E_l(z)$ are the Z-transforms of the input and the quantizer error introduced by the l^{th} stage. Note that our notation differs from that in [8] in that neither x nor e_l is normalized to unity. Thus, $x, e_l \in \{0, \dots, M-1\}$, rather than $\{0, \frac{(M-1)}{M}\}$, in this work. Furthermore, we consider the MASH 1-1-1 DDSM3 [12], the output of which is given by

$$Y(z) = \frac{1}{M}X(z) + \frac{1}{M}(1 - z^{-1})^3 E_3(z). \quad (3)$$

DDSMs are notorious for exhibiting spurs for particular values of their inputs and initial conditions. Certain combinations of inputs, initial conditions and bus widths produce short periodic cycles which in turn result in strong spurs [4]. Therefore, it is common practice to add a filtered dither signal to the input, as shown in Fig. 3. The resulting output is then

$$Y(z) = \frac{1}{M}X(z) + \frac{1}{M}(1 - z^{-1})^3 E_3(z) \frac{1}{M}V(z)D(z), \quad (4)$$

where $V(z)$ is of the form $(1 - z^{-1})^{l_d}$ with $l_d < l$ [7].

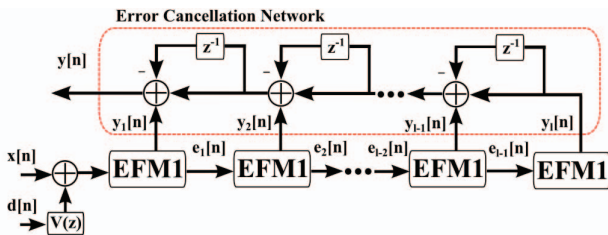


Fig. 3. DDSM with additive filtered dither

Fig. 4 shows the spectrum at the output of a dithered MASH 1-1-1 DDSM3. The envelope of the shaped quantization noise (black solid curve) is defined by

$$S_{DDSM3rd_{White}} = \frac{1}{12} \left(2 \sin \left(\frac{\pi f}{f_s} \right) \right)^6, \quad (5)$$

where $z = e^{j2\pi \frac{f}{f_s}}$.

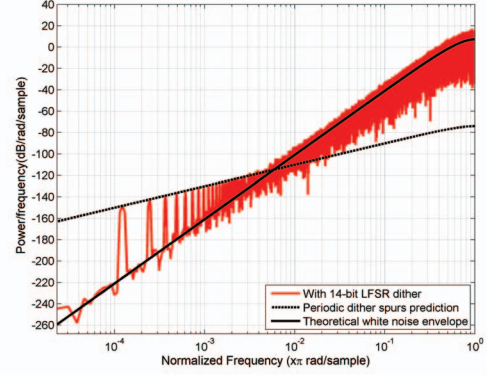


Fig. 4. Power spectral density of dithered MASH 1-1-1 DDSM3 with first-order shaped dither showing low frequency spurs caused by the LFSR.

At large values of the normalized frequency, the output (shown in red) is approximated well by the idealized third-order shaped white noise profile. At lower frequencies, the spectrum features spurs which are caused by the LFSR. These are bounded by the dashed black curve, the equation for which we will derive in the next section. In addition, we will show how to choose the wordlength of the LFSR to ensure that it does not produce any spurs in the output spectrum of the dithered DDSM.

III. PREDICTION OF DITHER SIGNAL'S SPUR HEIGHT

In this section, we derive an expression for the envelope of the dither-related spurs in Fig. 4. Recall Parseval's theorem, which states that the energy or power in the time-domain representation of a periodic signal with fundamental period L_s is equal to the energy or power in its frequency-domain representation, i.e.,

$$\frac{1}{L_s} \sum_{n=1}^{L_s} |x[n]|^2 = \sum_{k=1}^{L_s} |X[k]|^2, \quad (6)$$

where $x[n]$ is the discrete time series and $X[k]$ is its corresponding discrete-time Fourier series (DTFS).

In the case of the zeroth-order dithered MASH 1-1-1 with a maximum length PRBS generator, the length of the dither sequence $d[n] \in \{0, 1\}$ is $L_{sd} = 2^b - 1$ and its mean is

$$\bar{d}_b = \frac{2^{b-1}}{2^b - 1}. \quad (7)$$

Thus,

$$\frac{1}{L_{sd}} \sum_{n=1}^{L_{sd}} |d_e[n]|^2 = \sum_{k=1}^{L_{sd}} |D_e[k]|^2, \quad (8)$$

where and $d_e[n] = d[n] - \bar{d}$ is the DC-free signal. The total energy in one cycle of d_e is given by:

$$\sum_{n=1}^{L_{sd}} |d_e[n]|^2 = \frac{2^{b-2}}{(2^b - 1)^2}. \quad (9)$$

There are L_{sd} bins in the frequency range $[0, 2\pi)$ in an L_{sd} -point FFT. $L_{sd} - 1$ of these represent the signal power spectrum. We assume that the spectrum of the PRBS is white. Therefore, the power in each bin is given by:

$$S = \frac{1}{(L_{sd} - 1)L_{sd}} \sum_{n=1}^{L_{sd}} |d_e[n]|^2 = \frac{2^{b-2}}{(2^b - 1)^2}. \quad (10)$$

The processing gain PG of a window [13] is defined as the ratio of the output signal-to-noise ratio to the input signal-to-noise ratio. In particular, the processing gain of the Hanning window is given by $PG = (1/1.23)^2$.

Thus, the power per tone in the FFT is

$$S_{PSD} = \frac{2^{b-2}}{(2^b - 1)^2} \left(N_f PG \right). \quad (11)$$

The signal transfer function STF of the MASH 1-1-1 we consider in this paper is $STF = 1/M$. Therefore, the power spectral density of the zeroth order dither signal at the output of the DDSM is

$$S_0 = \frac{2^{b-2}}{(2^b - 1)^2} \left(\frac{N_f PG}{M^2} \right). \quad (12)$$

In general, the envelope of the LFSR-derived spurs in the case of l_d th order shaped additive LSB dither is defined by

$$S_{l_d} = S_0 \left(2 \sin \left(\frac{\pi f}{f_s} \right) \right)^{2l_d}. \quad (13)$$

Fig. 4 shows S_1 , resulting from first-order shaping, as a dashed curve.

Now that we know the relationship between the spur height profile and the number of bits in the LFSR, we can choose the LFSR's wordlength to hide the spurs under the envelope of the DDSM's shaped quantization noise.

IV. DDSM DITHERED BY PERIODIC PSEUDORANDOM NUMBER GENERATOR

In order to mask the dither signal, the fundamental frequency of the dither signal must be lower than the power spectral density of DDSM's shaped quantization noise at that frequency. The DDSM and pseudorandom number generator are finite state machines, so the amplitude of the power spectral density of DDSM and pseudorandom number generator depend on the number of points in the FFT (N_f), the period of the driven DDSM (L_{DDSM}) and the period of the pseudorandom number generator (L_{sd}). The spectral masking criterion is

$$S_0 \left(2 \sin \left(\frac{\pi f}{f_s} \right) \right)^{2l_d} < S_{DDSM_{Periodic}^{l_D}} @ f = f_s / L_{sd}. \quad (14)$$

$S_{DDSM_{Periodic}^{l_D}}$ is defined by

$$S_{DDSM_{Periodic}^{l_D}} = \frac{1}{12} \left(\frac{N_f PG}{L_{sd}} \right) \left(2 \sin \left(\frac{\pi f}{f_s} \right) \right)^{2l_D}, \quad (15)$$

where PG , l_d and l_D are the processing gain of the window, the order of dither filter and the order of the DDSM, respectively.

Substituting (12) and (15) into (14), and noting that

$$\sin(\pi f / f_s) \approx \pi f / f_s \quad \text{for } f \ll f_s, \quad (16)$$

we obtain

$$\frac{2^{b-2} N_f PG}{(2^b - 1)^2 M^2} < \frac{N_f PG}{12 \cdot 2(2^b - 1)M} \left(\frac{(2\pi)}{(2^b - 1)} \right)^{2(l_D - l_d)} \quad (17)$$

which reduces to

$$\log_2 \left(2^{b-2} (2^b - 1)^{2(l_D - l_d) - 1} \right) < N - \log_2 \left(\frac{24}{(2\pi)^{2(l_D - l_d)}} \right). \quad (18)$$

Noting that

$$2^{b-2} (2^b - 1)^{2(l_D - l_d) - 1} < 2^{2b(l_D - l_d) - 2}, \quad (19)$$

if we make

$$\log_2 \left(2^{2b(l_D - l_d) - 2} \right) < N - \log_2 \left(\frac{24}{(2\pi)^{2(l_D - l_d)}} \right), \quad (20)$$

then Eq. (18) will be satisfied.

This simplification gives the sufficient condition

$$b < \frac{N + 2 - \log_2 \left(\frac{24}{(2\pi)^{2(l_D - l_d)}} \right)}{2(l_D - l_d)}. \quad (21)$$

In order to ensure that L_{DDSM} is maximized, we choose

$$b = \left\lfloor \frac{N + 2 - \log_2 \left(\frac{24}{(2\pi)^{2(l_D - l_d)}} \right)}{2(l_D - l_d)} \right\rfloor, \quad (22)$$

where $\lfloor x \rfloor$ denotes the largest integer less than x .

V. DESIGN EXAMPLES

To illustrate the methodology, examples for second- and third-order MASH DDSMs with zeroth- and first-order shaped additive LSB dither are presented in this section.

A. Zeroth-order dithered MASH 1-1

In this example, shown in Fig. 5(a), each EFM1 in the MASH 1-1 has a 14-bit word. Substituting $N = 14$, $l_D = 2$ and $l_d = 0$ into (22) yields $b = 5$. Fig. 6(a) shows the simulated results. The simulated cycle length is $L_{DDSM} = 2 \times (2^5 - 1) \times 2^{14} = 1015808$; 2^{15} samples have been used to estimate the power spectral density.

B. First-order dithered MASH 1-1-1

With $N = 14$, $l_D = 3$ and $l_d = 1$, implemented in Fig. 5(b), Eq. (22) yields $b = 5$. Fig. 6(b) shows the simulated results. The cycle length is $L_{DDSM} = 2 \times (2^5 - 1) \times 2^{14} = 1015808$ and 2^{15} samples have again been used to estimate the power spectral density.

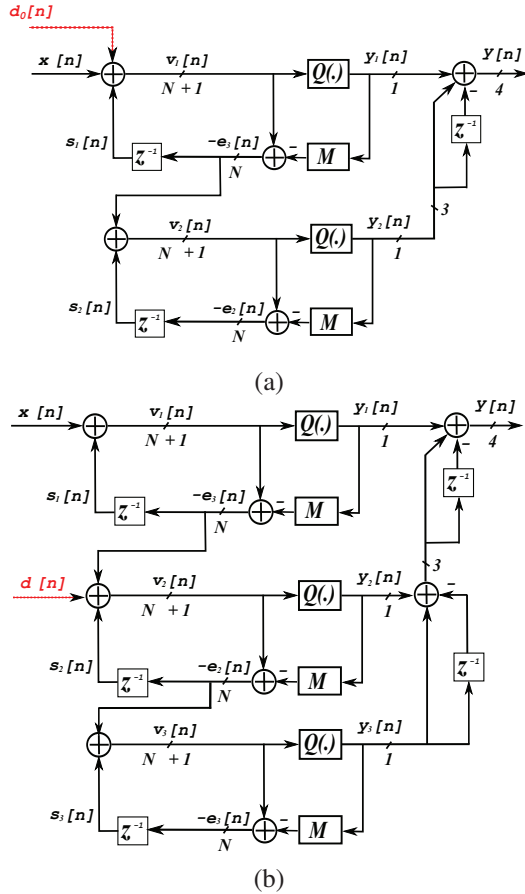


Fig. 5. Block diagrams of (a) the $MASH_{11}$ with zeroth-order dither and (b) the $MASH_{111}$ with first-order dither.

VI. CONCLUSION

When used to dither DDSMs, conventional LFSR-based PRBS generators can produce spurs. Pseudorandom dithering can be considered effective when the output spectrum of a DDSM contains, in addition to the desired signal spectrum, *only* noise which is closely approximated by shaped white quantization noise. In this work, we have presented a design equation (22) which defines the maximum size of LFSR to ensure effective (spur-free) dithering.

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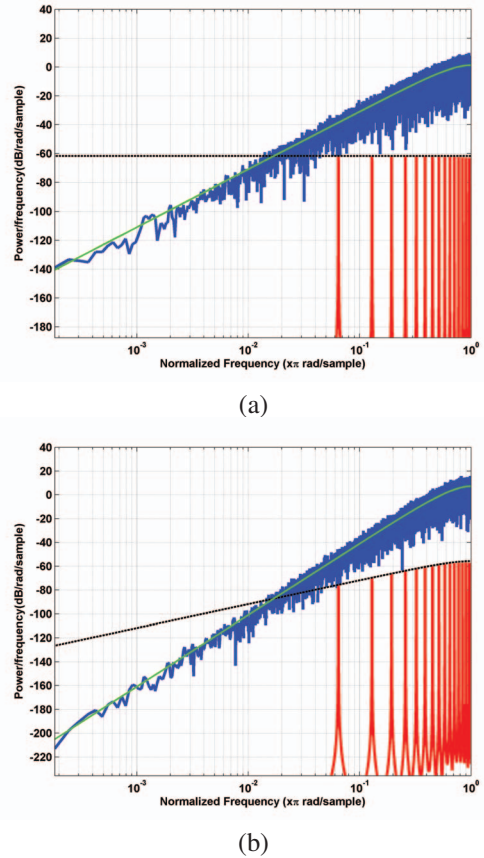


Fig. 6. Output phase noise spectra for (a) zeroth-order dithered MASH 1-1, and (b) first-order dithered MASH 1-1-1 with minimum size LFSRs. The LFSR-derived tones are masked below the shaped quantization noise.

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